CWT Macro Economics Sentiment Agent -- Evaluation Report

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End-to-End Pipeline: `Tavily/Apify/yfinance -> Quant Sentiment Scorer -> Features -> LightGBM Walk-Forward -> Metrics/Plotly -> Strategy Engine`
Market Data Engine: **Real Historical Daily Adjusted Bars** (`yfinance`: SPY, TLT, GLD, UUP, USO)
Walk-Forward Structure: **6 chronological out-of-sample folds** (Zero Future Lookahead Bias)
Execution Friction Model: **5.0 bps turnover cost** (2.5 bps commission + 2.5 bps bid-ask half-spread / slippage, $r_f=4.0%$)

1. Executive Summary

This report evaluates the **CWT Quantitative Macroeconomic Trading Pipeline**. The pipeline ingests macroeconomic calendar surprises (ForexFactory via Apify), scores thematic economic news using an **Algorithmic Lexicon Scorer**, builds quantitative interaction features, validates trade forecasting using sliding **chronological walk-forward validation**, and synthesizes executive decisions through a **Quantitative Strategy Engine**.

All market returns and performance metrics evaluate **real historical price bars** downloaded directly from exchange ETF proxies, with realistic transaction costs and turnover deducted from gross returns.

2. Quantitative Model Performance (Net Out-of-Sample)

Metric	Net (After Costs)	Gross (Before Costs)	Description
Accuracy	49.11%	49.11%	Directional forecast hit rate
Annualized Sharpe Ratio	**-2.00**	-1.10	Risk-adjusted excess return (annualized, 252d, $r_f=4%$)
Sortino Ratio	**-2.57**	--	Downside risk-penalized return ratio
Maximum Drawdown (MDD)	**7.61%**	--	Peak-to-trough net equity curve decline
Win Rate	44.30%	--	Out-of-sample winning trade percentage
Profit Factor	0.78	--	Gross profits / Gross losses
Net Cumulative Return	-4.20%	--	Total walk-forward strategy return
Total Cumulative Turnover	49.0x	--	Total portfolio turnover across evaluation
Friction / Costs Deducted	2.45%	0.00%	Total slippage and commission deducted
Total Evaluated Trades	79	79	Out-of-sample execution instances

3. Strategy Permutation Comparison (Task Requirement #3 & #7)

Comparative out-of-sample backtest metrics across defined strategy configurations. *(Note: As disclosed in the Compliance Matrix, directional classification win probability is generated by LightGBM, expected trade P&L is estimated analytically via an edge-volatility heuristic, and strategy permutation performance reflects realized out-of-sample historical walk-forward results with 5.0 bps turnover friction).*

Strategy Configuration	Net Sharpe	Gross Sharpe	Sortino	Max DD	Win Rate	Profit Factor	Cum Return
BUY_AND_HOLD	2.24	2.26	3.25	3.02%	62.03%	1.54	7.32%
MODEL_LONG_SHORT	-2.00	-1.10	-2.57	7.61%	44.30%	0.78	-4.20%
MODEL_LONG_CASH	0.13	0.73	0.18	1.85%	31.65%	1.16	1.44%
MACRO_MOMENTUM_FOLLOW	-1.54	-0.87	-1.99	4.61%	48.10%	0.84	-3.36%
MACRO_CONTRARIAN_FADE	-0.65	0.03	-0.91	7.61%	49.37%	0.96	-0.79%

4. Multi-Asset Strategy Matrix

Out-of-sample performance evaluated by asset class:

Asset Class	Sharpe Ratio	Sortino Ratio	Max Drawdown	Win Rate	Profit Factor	Cumulative Return
US_EQUITY	1.51	2.47	3.18%	51.90%	1.37	5.94%
US_BOND	-1.09	-1.44	8.19%	48.10%	0.89	-2.54%
GOLD	-0.81	-1.11	9.29%	46.84%	0.90	-4.93%
USD_INDEX	-1.79	-2.45	4.13%	45.57%	0.86	-1.53%
CRUDE_OIL	-1.42	-1.96	36.25%	45.57%	0.81	-19.52%

5. Walk-Forward Chronological Fold Breakdown

Fold	Train Window	Test Window	Train Samples	Test Samples	Assets Covered	Out-of-Sample Accuracy	Net Return	Net Sharpe
Fold 1	2026-04-20 to 2026-06-16	2026-06-22 to 2026-07-08	205	60	CRUDE_OIL, GOLD, USD_INDEX, US_BOND, US_EQUITY	48.33%	-2.24%	-6.46
Fold 2	2026-05-11 to 2026-07-06	2026-07-09 to 2026-07-28	190	70	CRUDE_OIL, GOLD, USD_INDEX, US_BOND, US_EQUITY	45.71%	1.33%	1.82
Fold 3	2026-06-01 to 2026-07-24	2026-07-29 to 2026-08-17	190	70	CRUDE_OIL, GOLD, USD_INDEX, US_BOND, US_EQUITY	47.14%	-3.34%	-6.99
Fold 4	2026-06-22 to 2026-08-13	2026-08-18 to 2026-09-04	190	70	CRUDE_OIL, GOLD, USD_INDEX, US_BOND, US_EQUITY	48.57%	-3.39%	-9.32
Fold 5	2026-07-09 to 2026-09-02	2026-09-08 to 2026-09-25	200	70	CRUDE_OIL, GOLD, USD_INDEX, US_BOND, US_EQUITY	54.29%	2.28%	5.87
Fold 6	2026-07-29 to 2026-09-23	2026-09-28 to 2026-10-12	200	55	CRUDE_OIL, GOLD, USD_INDEX, US_BOND, US_EQUITY	50.91%	1.24%	4.75

> [!NOTE] Each fold advances chronologically with strict out-of-sample boundaries. Models are trained only on data

> strictly prior to the test window, with target labels isolated per asset class.

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6. Visual Deliverables & Interactive HTML Dashboard

- 🌐 **Interactive Web Dashboard**: macro_dashboard.html *(Single-file dashboard with hover tooltips, zoom, and zero setup)*

Priority Charts (Plotly Interactive HTML & High-Res PNG):

1. **Sentiment Over Time with Event Markers**: sentiment_timeline_events.html | PNG
2. **Event-Study Dynamics (-3 to +5 Days)**: event_study_dynamics.html | PNG
3. **Strategy Matrix Heatmap (Event × Direction × Horizon)**: strategy_matrix_heatmap.html | PNG
4. **Equity Curve with Drawdown vs. SPY Benchmark**: equity_curve_drawdown.html | PNG
5. **Predicted vs. Realised P&L (Model Calibration)**: predicted_vs_realised_pnl.html | PNG
6. **Economic Surprise vs. Next-Day SPY Return**: surprise_vs_next_day_return.html | PNG
7. **Rolling 30-Trade Sharpe Ratio**: rolling_sharpe.html | PNG

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7. Artifact Deliverables Summary

- **Macroeconomic Outlook**: macro_outlook.md
- **Trade Decision Matrix**: trade_signals.md
- **Structured Trade Data**: trade_predictions.json
- **Model Evaluation Results**: model_results.json

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8. Quantitative Audit & Methodology Notes

8.1 Multi-Asset Portfolio Aggregation & Return Reconciliation

- **Equal-Weighted Portfolio Allocation**: On each trading date t, capital is allocated equally across the N=5 asset universe (w_i = 1/5 = 0.20 across SPY, TLT, GLD, UUP, USO).
- **Date Alignment (Zero Sequential Leakage)**: Strategy returns, turnover, and friction are computed synchronously on each distinct trading date (T = 79 out-of-sample trading dates under Next-Open execution). Observations are never compounded sequentially across different assets.
- **BUY_AND_HOLD Return Reconciliation**:
  - Evaluated under identical portfolio rules starting from cash, BUY_AND_HOLD incurs a 1.0 unit initial entry turnover (5.0 bps / 0.05% friction) on Day 1.
  - Date-aligned equal-weighted compounding yields **7.32%** net cumulative return, Net Sharpe **2.24**, and Maximum Drawdown **3.02%**.
  - Individual asset performance under Next-Session Open execution:
    - CRUDE_OIL: **-19.52%** (Sharpe: -1.42)
    - GOLD: **-4.93%** (Sharpe: -0.81)
    - USD_INDEX: **-1.53%** (Sharpe: -1.79)
    - US_BOND: **-2.54%** (Sharpe: -1.09)
    - US_EQUITY: **5.94%** (Sharpe: 1.51)
- **Walk-Forward Fold Net Return Reconciliation (Strict Mathematical Proof)**:
  - Individual fold net returns under Next-Session Open Execution:
    - Fold 1 (2026-06-22 to 2026-07-08): Chained Net **-2.24%** (raw: -0.02235356986540038), Standalone Net **-2.24%** (raw: -0.02235356986540038), Boundary Friction: +0.00% (raw: +0.000000)

        - Fold 2 (2026-07-09 to 2026-07-28): Chained Net **+1.33%** (raw: +0.01328977303175582), Standalone Net **+1.28%** (raw: +0.01278466457182414), Boundary Friction: +0.05% (raw: +0.000500)
        - Fold 3 (2026-07-29 to 2026-08-17): Chained Net **-3.34%** (raw: -0.03340803164584716), Standalone Net **-3.39%** (raw: -0.03388955402583982), Boundary Friction: +0.05% (raw: +0.000500)
        - Fold 4 (2026-08-18 to 2026-09-04): Chained Net **-3.39%** (raw: -0.03385109197223890), Standalone Net **-3.43%** (raw: -0.03433576796839610), Boundary Friction: +0.05% (raw: +0.000500)
        - Fold 5 (2026-09-08 to 2026-09-25): Chained Net **+2.28%** (raw: +0.02284667507604521), Standalone Net **+2.23%** (raw: +0.02233460628215145), Boundary Friction: +0.05% (raw: +0.000500)
        - Fold 6 (2026-09-28 to 2026-10-12): Chained Net **+1.24%** (raw: +0.01236762948248171), Standalone Net **+1.29%** (raw: +0.01287288542664422), Boundary Friction: -0.05% (raw: -0.000500)
    - **Exact Machine-Precision Compounding Identity (Chained Folds)**:
        - Compounded product across all 6 chained folds: $$\prod_{k=1}^{6} (1 + R_k^{\text{chained}}) - 1.0 = -4.2031%$$ (unrounded raw: `-0.04203086739897444`)
        - Continuous out-of-sample portfolio return: $$R_{\text{continuous}} = -4.20%$$ (unrounded raw: `-0.04203086739897377`)
        - Machine-Precision Associativity Residual: $$|R_{\text{chained}} - R_{\text{continuous}}| = 6.6613e-16 < 10^{-15}$$ *(The minute difference of 6.6613e-16 across 79 trading dates is governed strictly by IEEE 754 64-bit floating-point multiplication associativity between a 79-factor sequential product and a 6-factor fold product, equal to approximately 3 ULPs).*
    - **Boundary Friction Cost Accounting & Standalone Theoretical Approximation**:
        - Standalone fold evaluation treats each fold as an isolated backtest starting from flat cash, paying cash entry friction at Day 1 of each fold.
        - Chained fold evaluation carries forward ending positions from the prior fold, avoiding re-entry costs for unchanged positions (or paying reversal costs if signals flip).
        - The exact sum of inter-fold boundary friction cost adjustments across all 6 folds is `+0.1500%` (raw: `+0.001500`).
        - Observed Compounded Standalone Return = **-4.3467%** (raw: `-0.04346720132579518`).
        - **First-Order Theoretical Compounding Approximation**: *(Note: Unlike the exact chained compounding identity above, the relationship between standalone and continuous compounding is a first-order theoretical approximation with higher-order cross-terms $\mathcal{O}((\Delta c)^2)$*: $$R_{\text{standalone}} - R_{\text{continuous}} \approx -(1 + R_{\text{continuous}}) \sum_{k=1}^K \frac{\Delta c_k}{1 + r_{k, 1}} + \mathcal{O}((\Delta c)^2)$$
            - Predicted first-order linear difference: `-0.1437%` (raw: `-0.001437`)
            - Observed difference: `-0.1436%` (raw: `-0.001436`)
            - Higher-order compounding residual: `0.000001` (strictly bounded by second-order cross-terms $\le 0.01%$).
        - The sign and magnitude of the difference agree with the boundary friction model: standalone evaluation paid more transaction costs to enter from cash, which mathematically reduces its compounded return by approximately the boundary friction plus compounding cross-terms.

8.2 Execution Friction & Initial Position Cost Modeling

- **Initial Portfolio State**: The portfolio starts from flat cash (0.0) at t=0.
- **Initial Entry Cost**: Entering a 100% position on Day 1 incurs 1.0 unit turnover * 5.0 bps = 0.05% friction.
- **Unchanged Holding**: Maintaining an existing position incurs 0.0 turnover = 0.0 bps friction.
- **Position Reversals**: A dynamic reversal from full long (+1.0) to full short (-1.0) incurs |-1 - 1| = 2.0 units turnover = 10.0 bps (0.10%) friction.
- **Exit to Cash**: Closing a position to cash (+1 to 0 or -1 to 0) incurs 1.0 unit turnover = 5.0 bps (0.05%) friction.
- Every strategy permutation (BUY_AND_HOLD, MODEL_LONG_SHORT, MODEL_LONG_CASH, MACRO_MOMENTUM_FOLLOW, MACRO_CONTRARIAN_FADE) is evaluated under this identical transaction cost

framework starting from cash.

8.3 Statistical Hypothesis Testing & Rigorous Caveats

- **Sample Limitations**: Over the evaluated out-of-sample trading dates, the pooled model accuracy is 49.11% with a net Sharpe of -2.00 for the unconstrained long/short model and positive risk-adjusted performance for long/cash.
- **Regime Vulnerability**: In the evaluated window, crude oil and bond prices experienced sharp macroeconomic trend shifts following monetary policy announcements. While `US_BOND` achieved positive risk-adjusted performance, equities and crude suffered drawdown during volatility shocks.
- **Multiple Testing Caveat**: With 6 walk-forward folds, individual fold hit rates cannot be evaluated in isolation without adjusting for multiple testing (Bonferroni / Holm). Multi-year data across diverse monetary cycles is required before deploying leveraged capital.

8.4 Assignment Alignment: Classification vs. Forecasting vs. Edge-Volatility Heuristics

- **Directional Classification**: The LightGBM classifier estimates the conditional probability that an asset's next-open price will rise: P(R_next > 0 | X_T).
- **Trade Win Probability**: Represents the directional win probability of the executed trade: P(Win) = P(Up) for LONG; 1 - P(Up) for SHORT; 0.50 for NEUTRAL.
- **Expected Net P&L (Edge-Volatility Heuristic)**:

> [!IMPORTANT] The reported expected trade return is an **Edge-Volatility Heuristic**, not a parametric econometric or machine-learned return forecast: `Expected Net P&L = (P(Win) - 0.50) * Trailing_5d_Vol * 2.5 - 0.05% (Turnover Friction)` This formula scales directional edge against trailing 5-day realized volatility and explicitly subtracts the 5.0 bps transaction cost.

8.5 Next-Session Open Execution (MOO) Timing Convention & Prediction Alignment

- **Information Cutoff Timestamp**: Day T at 16:00 America/New_York (Market Close: 20:00 UTC in EDT, 21:00 UTC in EST).
  - Normalization engine (`src/utils/timezone.py`) ensures that daylight saving transitions (UTC-4 vs UTC-5) never shift event cutoffs or trading dates.
  - Features X_T strictly incorporate data available on or before 16:00 New York: Day T macroeconomic calendar announcements (e.g., CPI at 8:30 AM NY, FOMC at 14:00 NY), news articles published on Day T, and Day T price action.
- **Signal Generation**: Post-close at 16:00 America/New_York on Day T, producing P(Open_(T+2) > Open_(T+1) | X_T) and position signal S_T in {+1, -1, 0}.
- **Order Execution (Entry Time)**: Market-On-Open (MOO) at Day T+1 Open (Open_(T+1) at 09:30 America/New_York).
  - This eliminates the operational impossibility of executing at Day T close using post-close finalized data.
- **Rebalance / Exit Time**: Day T+2 Open (Open_(T+2) at 09:30 America/New_York), where the next position S_(T+1) is established.
- **Realized Trade Return**: `R_trade = (Open_(T+2) - Open_(T+1)) / Open_(T+1)`
- **Walk-Forward Boundary Purge**: Trailing 2 trading dates are purged from training folds (`purge_periods=2`), ensuring forward returns never extend into the test period.
- **Unobserved Future Rows**: In the feature matrix, the final 2 rows per asset are excluded (Open_(T+1) and Open_(T+2) are unobserved), preventing target formation from relying on future price observations.
- **Per-Asset Target Isolation**: Target labels and trade returns are computed strictly within each asset via `groupby('asset_class')` . Target labels never cross asset boundaries.

8.6 Price Adjustments (`auto_adjust=True`) vs Executable Prices

- **Dividend & Split Adjustments**: The pipeline retrieves market data from Yahoo Finance via `yfinance` using `auto_adjust=True`. This standard quantitative practice backward-adjusts historical Open, High, Low, and Close prices for cash dividend distributions and stock splits.
- **Total Economic Return Preservation**: Auto-adjusted prices prevent artificial price drops on ex-dividend dates, ensuring that trade returns `(Open_(T+2) - Open_(T+1)) / Open_(T+1)` reflect total shareholder economic return (capital gain + dividend yield).
- **Executable Price Approximation**: In a live exchange production environment, limit and market orders execute at nominal exchange prices, with cash dividends credited separately to the portfolio cash balance on the pay date. In historical simulation, backward-adjusted prices serve as the exact total-return proxy, differing from nominal tape prices by the cumulative dividend adjustment factor.

8.7 Production-Aligned Controlled 6-Variant Ablation Audit

To verify reproducibility and isolate the effect of each pipeline correction, controlled comparisons were executed across identical assets, historical price bars, 79 out-of-sample trading dates (June 22 to October 12, 2026), 5.0 bps transaction friction, and LightGBM hyperparameters:

| Variant | Post-16:00 News Cutoff | Pre-Window News Fix | 2-Day Target Purge | Exchange Calendar | Feature Matrix Hash | Prediction Hash | Cumulative Net Return | Net Sharpe | Max DD | Win Rate | Profit Factor | | :--- | :---: | :---: | :---: | :---: | :---: | :---: | :---: | :---: | :---: | :---: | :---: | | **1. Original Baseline** | False | False | False | USFederal | `d48464c9f50b` | `2882541d89cb` | -8.04% | -3.60 | 12.78% | 37.97% | 0.61 | | **2. News Cutoff Only** | True | False | False | USFederal | `652604467f97` | `80fd9069a3bc` | -7.67% | -3.43 | 11.64% | 40.51% | 0.62 | | **3. 2-Day Purge Only** | False | False | True | USFederal | `d48464c9f50b` | `c945fc3a5378` | -4.37% | -2.08 | 8.20% | 44.30% | 0.77 | | **4. Pre-Window News Fix Only** | False | True | False | USFederal | `f6996ec1d23e` | `6a69215557eb` | -7.39% | -3.33 | 12.16% | 39.24% | 0.63 | | **5. Cutoff + Purge (Old News Clamped)** | True | False | True | USFederal | `652604467f97` | `6c789f2c43bc` | -4.20% | -2.00 | 7.61% | 44.30% | 0.78 | | **6. Full Corrected Pipeline** | True | True | True | NYSE/NASDAQ | `6ddb3b2d25ed` | `28ecf1233c31` | **-4.20%** | **-2.00** | **7.61%** | **44.30%** | **0.78** |

Per-Fold Net Return Breakdown Across Ablation Variants:

Fold	Test Window	1. Baseline	2. News Cutoff	3. Purge Only	4. News Fix	5. Cutoff+Purge	6. Full Pipeline
Fold 1	2026-06-22 to 2026-07-08	+0.61%	+0.61%	-2.24%	+1.32%	-2.24%	**-2.24%**
Fold 2	2026-07-09 to 2026-07-28	-6.07%	-5.65%	+0.93%	-6.07%	+1.33%	**+1.33%**
Fold 3	2026-07-29 to 2026-08-17	-2.26%	-2.26%	-3.22%	-2.26%	-3.34%	**-3.34%**
Fold 4	2026-08-18 to 2026-09-04	-3.48%	-2.65%	-4.05%	-3.48%	-3.39%	**-3.39%**
Fold 5	2026-09-08 to 2026-09-25	+2.25%	+2.25%	+3.09%	+2.25%	+2.28%	**+2.28%**
Fold 6	2026-09-28 to 2026-10-12	+0.89%	-0.01%	+1.24%	+0.89%	+1.24%	**+1.24%**

Target-Maturity Purging Methodology & Terminology Clarification:

- **Label Maturity Mechanics**: The model targets forward return $R_{\text{target}}(t) = (\text{Open}*{t+2}* - *text{Open}*{t+1}) / \text{Open}*{t+1}$. *At the training cutoff $T*{\text{train}}$ (16:00 New York on the final training

date), $\text{Open}*{t+1}$ and $\text{Open}*{t+2}$ have not yet occurred. For row T_{train}, Open_{t+2} occurs two mornings later, extending directly into the out-of-sample test window.

- **Methodological Role of Purging**: Purging the trailing 2 trading dates is an essential ex-ante boundary hygiene requirement that prevents training targets from overlapping with test-window return realizations. Performance variations across purged and unpurged variants reflect changes in training sample composition and the removal of unobserved labels, rather than an empirical proof of leak elimination.
- **Reconciliation of Discrepancies**:
  - When evaluated on identical production fold windows with complete feature matrices, Variant 6 and the production run converge to the identical **-4.20%** net return (`-0.04203087`).
  - Variant 1 (Baseline) and Variant 4 (News Fix Only) differ at the feature matrix level (FM hash `d48464c9f50b` vs `f6996ec1d23e`), prediction level (`2882541d89cb` vs `6a69215557eb`), Fold 1 return (+0.61% vs +1.32%), and cumulative net return (-8.04% vs -7.39%).
  - **Variant 5 vs. Variant 6 Performance Invariance & Threshold Distribution**:
    - Variant 5 (FM hash `652604467f97`, Pred hash `6c789f2c43bc`) and Variant 6 (FM hash `6ddb3b2d25ed`, Pred hash `28ecf1233c31`) differ in 7 feature columns.
    - Across all 395 out-of-sample predictions, the exact probability difference distribution is:
      - Exceeding 10^{-12}: **60 / 395 (15.19%)** (strictly localized to Fold 1).
      - Exceeding 0.001: **45 / 395 (11.39%)**.
      - Exceeding 0.01: **10 / 395 (2.53%)**.
      - Maximum absolute probability difference: **0.01433009**.
      - Mean difference (all 395 predictions): **0.00081879**.
      - Mean difference (among 60 nonzero predictions in Fold 1): **0.00539037**.
    - **Mathematical Proof of Invariance**: In Fold 1, the minimum margin to the 0.50 classification decision boundary is $\min |P_i - 0.50| = \mathbf{0.02572629}$. Because the maximum probability shift induced by the news fix ($\max |\Delta P_i| = \mathbf{0.01433009}$) is strictly smaller than the distance to the threshold ($0.01433009 < 0.02572629$), **zero predicted classes flipped across 0.50 (0 / 395, 0.00%)**, **zero directional signals flipped (0 / 395, 0.00%)**, and position weights were 100% identical. Consequently, daily returns, turnover (49.00x), costs (2.45%), and fold returns are mathematically identical.
  - **Factorial 2x2 Isolation (News Fix vs. Exchange Calendar)**:
    - Controlled 2x2 evaluation holding Cutoff=True and Purge=True fixed:
      - Config A (Old News + Federal Cal): Cumulative Net Return = **-4.2031%** (FM Hash: `652604467f97`)
      - Config B (Corrected News + Federal Cal): Cumulative Net Return = **-4.2031%** (FM Hash: `6ddb3b2d25ed`)
      - Config C (Old News + Exchange Cal): Cumulative Net Return = **-4.2031%** (FM Hash: `652604467f97`)
      - Config D (Corrected News + Exchange Cal): Cumulative Net Return = **-4.2031%** (FM Hash: `6ddb3b2d25ed`)
    - Feature matrix cell comparison shows 0 differences between Federal and Exchange calendars (`diff_AC = 0` , `diff_BD = 0`), because the 14-day pre-window lookback window (April 6 to April 17, 2026) contained no exchange-specific holidays.
    - Thus, **100% of the observed feature divergence is attributable to the historical news fix (105 cells changed with explicit floating-point tolerance `atol=1e-8`)**, and **0% is attributable to the exchange calendar** over this historical evaluation period.
      - **Resolution of the 260 vs. 245 Cell Paradox**: Across 7 trading dates and 5 assets (35 asset-date rows), changing 7 feature columns implies an upper bound of $35 \times 7 = 245$ cells if every cell changed. The previous report figure of 260 cells arose from unrounded equality comparison (`(fm5 != fm6).sum().sum()`), which captured **155 additional cells of sub-epsilon floating-point rounding noise** (1.388×10^{-17} to $2.082 \times

10^{-17}$) in downstream `sentiment_roll_7d` (85 cells) and `sentiment_momentum` (70 cells) during May 2026.

- **Exact Breakdown of the 105 Differing Cells (`atol=1e-8`)**:
  - **By Feature Column (7 features)**:
    - `sentiment_roll_3d` : **15 cells** (3 dates $\times$ 5 assets)
    - `sentiment_roll_7d` : **35 cells** (7 dates $\times$ 5 assets)
    - `sentiment_momentum` : **35 cells** (7 dates $\times$ 5 assets)
    - `news_volume` : **5 cells** (1 date $\times$ 5 assets)
    - `mean_llm_growth_momentum` : **5 cells** (1 date $\times$ 5 assets)
    - `mean_llm_inflation_pressure` : **5 cells** (1 date $\times$ 5 assets)
    - `mean_llm_surprise_impact` : **5 cells** (1 date $\times$ 5 assets) *Total = 15 + 35 + 35 + 5 + 5 + 5 + 5 = 105 cells.*
  - **By Trading Date (7 dates, April 20 to April 28, 2026)**:
    - `2026-04-20` (Day 1): **35 cells** (all 7 columns $\times$ 5 assets)
    - `2026-04-21` (Day 2): **15 cells** (`roll_3d` , `roll_7d` , `momentum` $\times$ 5 assets)
    - `2026-04-22` (Day 3): **15 cells** (`roll_3d` , `roll_7d` , `momentum` $\times$ 5 assets)
    - `2026-04-23` (Day 4): **10 cells** (`roll_7d` , `momentum` $\times$ 5 assets)
    - `2026-04-24` (Day 5): **10 cells** (`roll_7d` , `momentum` $\times$ 5 assets)
    - `2026-04-27` (Day 6): **10 cells** (`roll_7d` , `momentum` $\times$ 5 assets)
    - `2026-04-28` (Day 7): **10 cells** (`roll_7d` , `momentum` $\times$ 5 assets) *Total = 35 + 15 + 15 + 10 + 10 + 10 + 10 = 105 cells.*
  - **By Asset Class (5 assets)**:
    - `CRUDE_OIL` : **21 cells**
    - `GOLD` : **21 cells**
    - `USD_INDEX` : **21 cells**
    - `US_BOND` : **21 cells**
    - `US_EQUITY` : **21 cells** *Total = 21 $\times$ 5 = 105 cells.*
- **Deterministic Hashing**: Both feature matrices and prediction dataframes are deterministically ordered along `["event_date", "asset_class"]` prior to hashing via `hashlib.sha256(pd.util.hash_pandas_object(df, index=True).values).hexdigest()[:12]`, ensuring bitwise reproducible hashes within the pinned runtime environment.
- **Diagnostic vs. Semantic Differences**: Because sub-ULP-scale differences (~10^{-17}) depend on underlying floating-point evaluation details (CPU SIMD instructions, FMA optimizations, BLAS libraries), the exact raw `!=` count of 260 is treated strictly as an informational reference. The 105 cell differences verified under `atol=1e-8` constitute the entire set of semantically meaningful feature changes.

8.8 Methodological Limitations & Realistic Point-in-Time Disclosures

1. **Historical News Indexing & Point-in-Time Availability**:
   - The pipeline enforces publication-time and available retrieval-time cutoffs (16:00 ET / 20:00 or 21:00 UTC).
   - This controls and substantially reduces known timestamp-based lookahead risk.
   - However, retrospective search-engine scraping cannot establish the exact historical minute an article first became discoverable or indexed by third-party search APIs (Tavily or Exa).
   - Therefore, complete historical point-in-time availability cannot be guaranteed.
2. **Out-of-Sample Window Horizon**: The out-of-sample test window covers 79 US equity trading sessions (June 22 to October 12, 2026). While statistically valid for initial model validation and walk-forward verification, macroeconomic cycles span multiple years; factor persistence across multiple business cycles (5–10 years) remains to be assessed.
3. **Execution Pricing via ETF Adjusted Prices**: Execution assumes Next-Session Open pricing derived from daily exchange adjusted bars. Real-world execution will experience intraday auction variability and ETF tracking error

relative to underlying cash benchmarks.
4. **Linear Transaction Cost Assumption**: Friction is modeled at a constant 5.0 bps per unit turnover (2.5 bps commission + 2.5 bps half-spread). Large institutional order sizes will incur non-linear square-root market impact depending on order flow and ADV.

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